forward process:

Random walk away from image.

Image : x_0

Markov process: $\boxed{q(x_t \mid x_{t-1}) = N(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I)}$

$$\Rightarrow \boxed{\begin{aligned} x_t &= \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\, \epsilon \\ \epsilon &\sim N(0,1) \end{aligned}}$$

$$\Rightarrow x_1 = \sqrt{1-\beta_1}\, x_0 + \sqrt{\beta_1}\, \epsilon$$

$$x_2 = \sqrt{1-\beta_2}\, x_1 + \sqrt{\beta_2}\, \epsilon$$

$$x_3 = \sqrt{1-\beta_3}\, x_2 + \sqrt{\beta_3}\, \epsilon$$

$$\vdots$$

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\, \epsilon$$

$$\Rightarrow x_2 = \sqrt{1-\beta_2}\left(\sqrt{1-\beta_1}\, x_0 + \sqrt{\beta_1}\, \epsilon\right) + \sqrt{\beta_2}\, \epsilon$$

$$= \sqrt{1-\beta_2}\sqrt{1-\beta_1}\, x_0 + \sqrt{1-\beta_2}\sqrt{\beta_1}\, \epsilon + \sqrt{\beta_2}\, \epsilon$$

$$x_3 = \sqrt{1-\beta_3}\left(\sqrt{1-\beta_2}\sqrt{1-\beta_1}\, x_0 + \sqrt{1-\beta_2}\sqrt{\beta_1}\, \epsilon + \sqrt{\beta_2}\, \epsilon\right) + \sqrt{\beta_3}\, \epsilon$$

$$= \sqrt{1-\beta_3}\sqrt{1-\beta_2}\sqrt{1-\beta_1}\, x_0 + \sqrt{1-\beta_3}\sqrt{1-\beta_2}\sqrt{\beta_1}\, \epsilon + \sqrt{1-\beta_3}\sqrt{\beta_2}\, \epsilon + \sqrt{\beta_3}\, \epsilon$$

$$\Rightarrow x_t = \prod_{s=1}^{t} \sqrt{1-\beta_s}\, x_0 + \sum_{s=1}^{t} \sqrt{\beta_s} \prod_{j=s+1}^{t} \sqrt{1-\beta_j}\, \epsilon$$

let $\gamma_s = \sqrt{\beta_s} \prod_{j=s+1}^{t} \sqrt{1-\beta_j}$

$$\Rightarrow x_t = \prod_{s=1}^{t} \sqrt{1-\beta_s}\, x_0 + \sum_{s=1}^{t} \gamma_s \epsilon$$

let $\alpha_t = 1-\beta_t$

$$\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$$

Note $\epsilon_1 = N(\mu_1, \sigma_1^2)$

$\epsilon_2 = N(\mu_2, \sigma_2^2)$

$\epsilon_1 + \epsilon_2 = N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

$$\Rightarrow x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sum_{s=1}^{t} \gamma_s N(0,1)$$

$$\Rightarrow x_t = \sqrt{\bar{\alpha}_t}\, x_0 + N\left(0, \sum_{s=1}^{t} \gamma_s^2\right)$$

$$\hookrightarrow \sum_{s=1}^{t} \gamma_s^2 = \sum_{s=1}^{t} \beta_s \prod_{j=s+1}^{t} 1-\beta_j$$

$\beta_s = 1-(1-\beta_s)$

$$= \sum_{s=1}^{t} \left[1-(1-\beta_s)\right] \prod_{j=s+1}^{t} 1-\beta_j$$

$$= \sum_{s=1}^{t} \prod_{j=s+1}^{t} 1-\beta_j - \prod_{j=s+1}^{t} (1-\beta_j)(1-\beta_s)$$

$$= \sum_{s=1}^{t} \prod_{j=s+1}^{t} 1-\beta_j - \frac{\prod_{j=s}^{t} (1-\beta_j)(1-\beta_s)}{(1-\beta_s)}$$

$$= \sum_{s=1}^{t} \prod_{j=s+1}^{t} 1-\beta_j - \prod_{j=s}^{t} 1-\beta_j$$

$s=1: (1-\beta_2)(1-\beta_3)\cancel{\cdots(1-\beta_t)} - (1-\beta_1)(1-\beta_2)\cdots(1-\beta_t)$

$s=2: (1-\beta_3)(1-\beta_4)\cancel{\cdots(1-\beta_t)} - (1-\beta_2)(1-\beta_3)\cancel{\cdots(1-\beta_t)}$

$s=3: (1-\beta_4)(1-\beta_5)\cdots(1-\beta_t) - (1-\beta_3)(1-\beta_4)\cancel{\cdots(1-\beta_t)}$

$\vdots$

$s=t: 1-(1-\beta_t)$

empty set goes to 1

$$= 1 - \prod_{i=1}^{t} 1-\beta_i$$

$$= 1 - \bar{\alpha}_t$$

$$\Rightarrow \sum_{s=1}^{t} \gamma_s^2 = 1-\bar{\alpha}_t$$

$$\Rightarrow \boxed{\begin{aligned} &x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon \\ &\text{with } \alpha_t = 1-\beta_t \\ &\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s \\ &\epsilon = \mathcal{N}(0,1) \end{aligned}}$$

Backwards Process:

$$P(A \mid B, C) = \frac{P(A,B,C)}{P(B,C)}$$

Chain rule

can neglect x_0 b/c markov process

$$P(x_{t-1} \mid x_0, x_t) = \frac{P(x_0, x_{t-1}, x_t)}{P(x_0, x_t)} = \frac{P(x_0)\, P(x_{t-1} \mid x_0)\, P(x_t \mid x_{t-1}, x_0)}{P(x_0, x_t)}$$

only care about terms with x_{t-1}

$$P(x_{t-1} \mid x_0, x_t) \propto P(x_{t-1} \mid x_0)\, P(x_t \mid x_{t-1})$$

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t ; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I\right)$$

$$= \frac{1}{(2\pi \beta_t)^{k/2}} e^{-\frac{1}{2\beta_t} \| x_t - \sqrt{1-\beta_t}\, x_{t-1} \|_k^2}$$

$$q(x_{t-1} \mid x_0) = \mathcal{N}\left(x_{t-1} ; \sqrt{\bar{\alpha}_{t-1}}\, x_0, 1-\bar{\alpha}_{t-1}\right)$$

$$= \frac{1}{(2\pi [1-\bar{\alpha}_{t-1}])^{k/2}} e^{\frac{-1}{2(1-\bar{\alpha}_{t-1})} \| x_{t-1} - \sqrt{\bar{\alpha}_{t-1}}\, x_0 \|_k^2}$$

$$q(x_{t-1} \mid x_0, x_t) \propto \exp\left[\frac{-1}{2\beta_t}\| x_t - \sqrt{1-\beta_t}\, x_{t-1}\|_k^2 - \frac{1}{2(1-\bar{\alpha}_{t-1})}\| x_{t-1} - \sqrt{\bar{\alpha}_{t-1}}\, x_0\|_k^2\right]$$

$$\propto \exp\Big[\frac{-1}{2\beta_t}\left(x_t^2 - 2\sqrt{1-\beta_t}\, x_t x_{t-1} + [1-\beta_t] x_{t-1}^2\right)$$

$$\frac{-1}{2(1-\bar{\alpha}_{t-1})}\left(x_{t-1}^2 - 2\sqrt{\bar{\alpha}_{t-1}}\, x_{t-1} x_0 + \bar{\alpha}_{t-1}^2 x_0^2\right)\Big]$$

$$\propto \exp\Big[\frac{\sqrt{1-\beta_t}\, x_t x_{t-1}}{\beta_t} - \frac{[1-\beta_t] x_{t-1}^2}{2\beta_t}$$

$$\frac{-x_{t-1}^2}{2(1-\bar{\alpha}_{t-1})} + \frac{2\sqrt{\bar{\alpha}_{t-1}}\, x_{t-1} x_0}{2(1-\bar{\alpha}_{t-1})} + C\Big]$$

$$\propto \exp\Big[\left(-\frac{[1-\beta_t]}{2\beta_t} - \frac{1}{2(1-\bar{\alpha}_{t-1})}\right) x_{t-1}^2$$

$$+\left(\frac{\sqrt{1-\beta_t}\, x_t}{\beta_t} + \frac{\sqrt{\bar{\alpha}_{t-1}}\, x_0}{1-\bar{\alpha}_{t-1}}\right) x_{t-1}\Big]$$

$$= \exp\Big[-\frac{1}{2}\Big[\left(\frac{[1-\beta_t]}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}}\right) x_{t-1}^2$$

$$+\left(\frac{-2\sqrt{1-\beta_t}\, x_t}{\beta_t} - \frac{2\sqrt{\bar{\alpha}_{t-1}}\, x_0}{1-\bar{\alpha}_{t-1}}\right) x_{t-1}\Big]\Big]$$

Want in form $\exp\left[-\frac{1}{2}\frac{\|x_{t-1} - \tilde{\mu}_t\|_k^2}{\tilde{\beta}_t}\right]$

let $a = \frac{[1-\beta_t]}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}}$

$b = \frac{-2\sqrt{1-\beta_t}\, x_t}{\beta_t} - \frac{2\sqrt{\bar{\alpha}_{t-1}}\, x_0}{1-\bar{\alpha}_{t-1}}$

complete the square

$$\Rightarrow \propto \exp\left[-\frac{1}{2}\left(a x_{t-1}^2 + b x_{t-1}\right)\right] = \exp\left[-\frac{1}{2} a\left(x_{t-1}^2 + \frac{b}{a} x_{t-1}\right)\right] = \exp\left[-\frac{1}{2} a\left(x_{t-1} + \frac{b}{2a}\right)^2\right]$$

$$\Rightarrow \quad \tilde{\beta}_t = \frac{1}{a} = \frac{1}{\left(\frac{1-\beta_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}}\right)}$$

$$\tilde{\mu}_t = \frac{-b}{2a} = \left(\frac{\sqrt{1-\beta_t}\, x_t}{\beta_t} + \frac{\sqrt{\bar{\alpha}_{t-1}}\, x_0}{1-\bar{\alpha}_{t-1}}\right)\tilde{\beta}_t$$

substituting in: $\alpha_t = 1-\beta_t$, $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_t$

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon \quad \Rightarrow \quad x_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(x_t - \sqrt{1-\bar{\alpha}_t}\, \epsilon_t\right)$$

$$\Rightarrow \quad \tilde{\beta}_t = \frac{1}{\frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}}} = \frac{1}{\left(\frac{\alpha_t(1-\bar{\alpha}_{t-1}) + \beta_t}{\beta_t(1-\bar{\alpha}_{t-1})}\right)} = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{\alpha_t(1-\bar{\alpha}_{t-1}) + \beta_t}$$

$\alpha_t\, \bar{\alpha}_{t-1} = \bar{\alpha}_t$

$$= \frac{\beta_t(1-\bar{\alpha}_{t-1})}{\alpha_t - \bar{\alpha}_t + \beta_t}$$

$1-\beta_t$

$$\boxed{\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t}$$

$$\Rightarrow \quad \tilde{\mu}_t = \left(\frac{\sqrt{\alpha_t}\, x_t}{\beta_t} + \frac{\sqrt{\bar{\alpha}_{t-1}}\, x_0}{1-\bar{\alpha}_{t-1}}\right)\left(\frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t\right)$$

$$= \frac{\sqrt{\alpha_t}\, x_t\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t} + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_t}\beta_t\, x_0$$

$$= \frac{\sqrt{\alpha_t}\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_t} \beta_t \left[\frac{1}{\sqrt{\bar{\alpha}_t}}\left(x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_t\right)\right]$$

$$= \frac{\alpha_t - \bar{\alpha}_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} x_t$$

$$= \frac{\sqrt{\alpha_t}\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t} x_t$$

$$+ \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_t} \frac{\beta_t}{\sqrt{\bar{\alpha}_t}} x_t - \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_t} \frac{\beta_t}{\sqrt{\bar{\alpha}_t}} \sqrt{1-\bar{\alpha}_t}\,\epsilon_t$$

$$= \frac{1-\alpha_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} x_t \qquad = \frac{1-\alpha_t}{1-\bar{\alpha}_t} \frac{1}{\sqrt{\alpha_t}} \sqrt{1-\bar{\alpha}_t}\,\epsilon_t$$

$$= \frac{1-\bar{\alpha}_t + \alpha_t - \alpha_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} x_t - \frac{1-\alpha_t}{\sqrt{\alpha_t}} \frac{\sqrt{1-\bar{\alpha}_t}}{1-\bar{\alpha}_t} \epsilon_t$$

$$= \frac{1}{\sqrt{\alpha_t}} x_t - \frac{1-\alpha_t}{\sqrt{\alpha_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_t$$

$$= \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_t\right)$$

$$\boxed{\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_t\right)}$$

Training loop:

Predict $\epsilon_\theta(x_t, t) \sim E[\epsilon | x_t]$ "mean noise added to reach x_t"

0) define β_t and precompute $\sqrt{\bar{\alpha}_t}$ & $\sqrt{1-\bar{\alpha}_t}$

$\alpha_t = 1 - \beta_t$

$\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$

1) Sample image (normalized $[-1, 1]$)

$x_0 \sim X$

2) Sample time step

$t \sim \{1, \ldots, T\}$

3) Sample noise

$\epsilon \sim N(0, I)$

4) Compute forward process

$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$

5) pass through network

$\epsilon_\theta(x_t, t)$

6) Compute loss

$L = \| \epsilon - \epsilon_\theta(x_t, t) \|^2$

7) update parameters

⋮

Sample Loop:

1) Generate Starting point

$$X_t = T = N(0,1)$$

2) predict noise

$$\hat{\epsilon}_\theta(x_t, t)$$

3) compute new mean, var, ϵ

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\hat{\epsilon}_\theta(x_t, t)\right)$$

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$$

$$\epsilon = N(0,1)$$

4) compute x_{t-1}

$$x_{t-1} = \tilde{\mu}_t + \sqrt{\tilde{\beta}_t}\,\epsilon$$

5) repeat until $t=1$

$$x_0 = \tilde{\mu}_t \quad \text{(no added noise)}$$